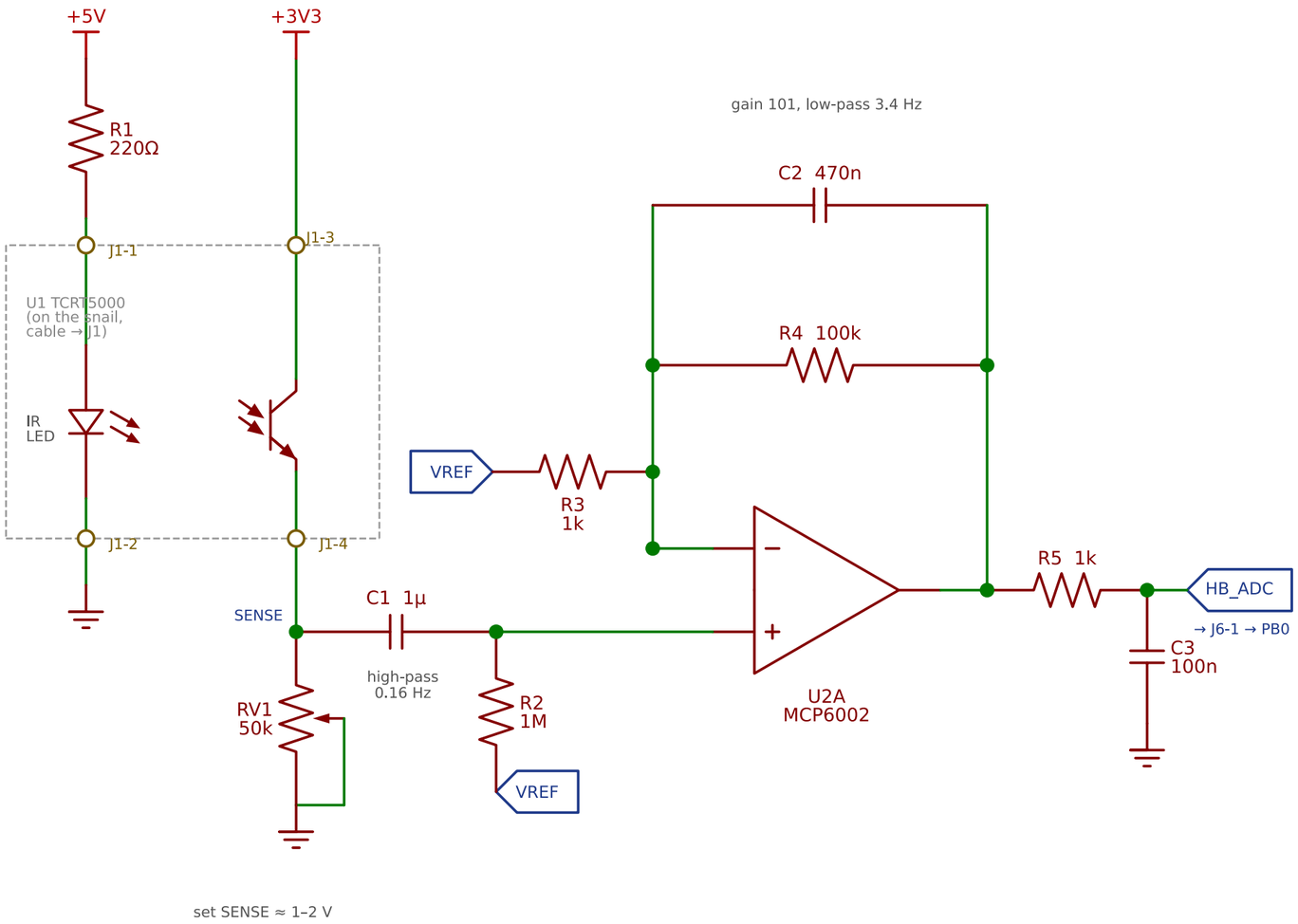
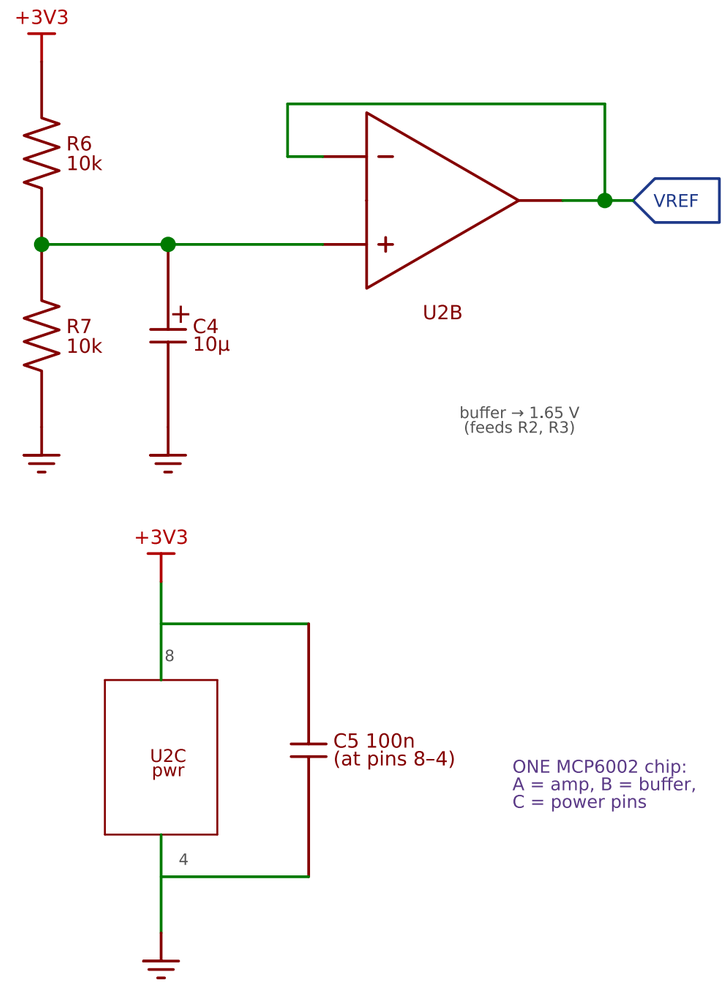
# SnailPulse — how to wire your Orbita26 schematic (refs match your KiCad sheet)

Read it like KiCad: wires inside a block are real wires; a label/power symbol with the same name anywhere = the same net.

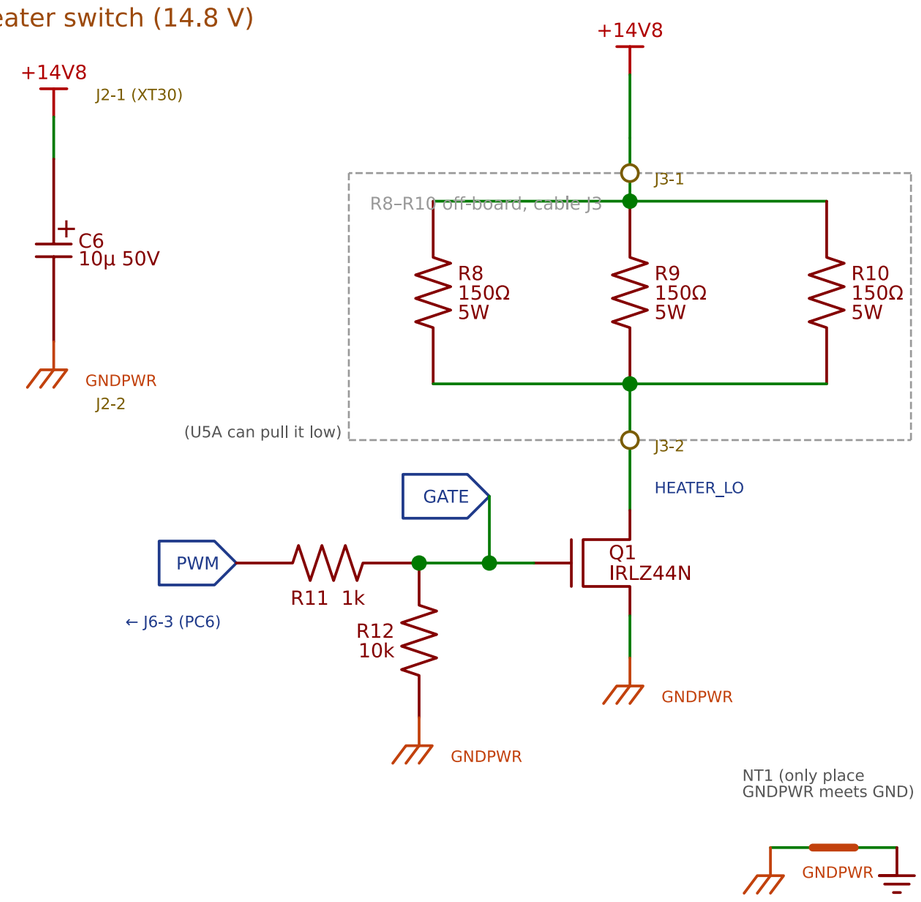
## A · Heartbeat sensor + amplifier → STM32 PB0



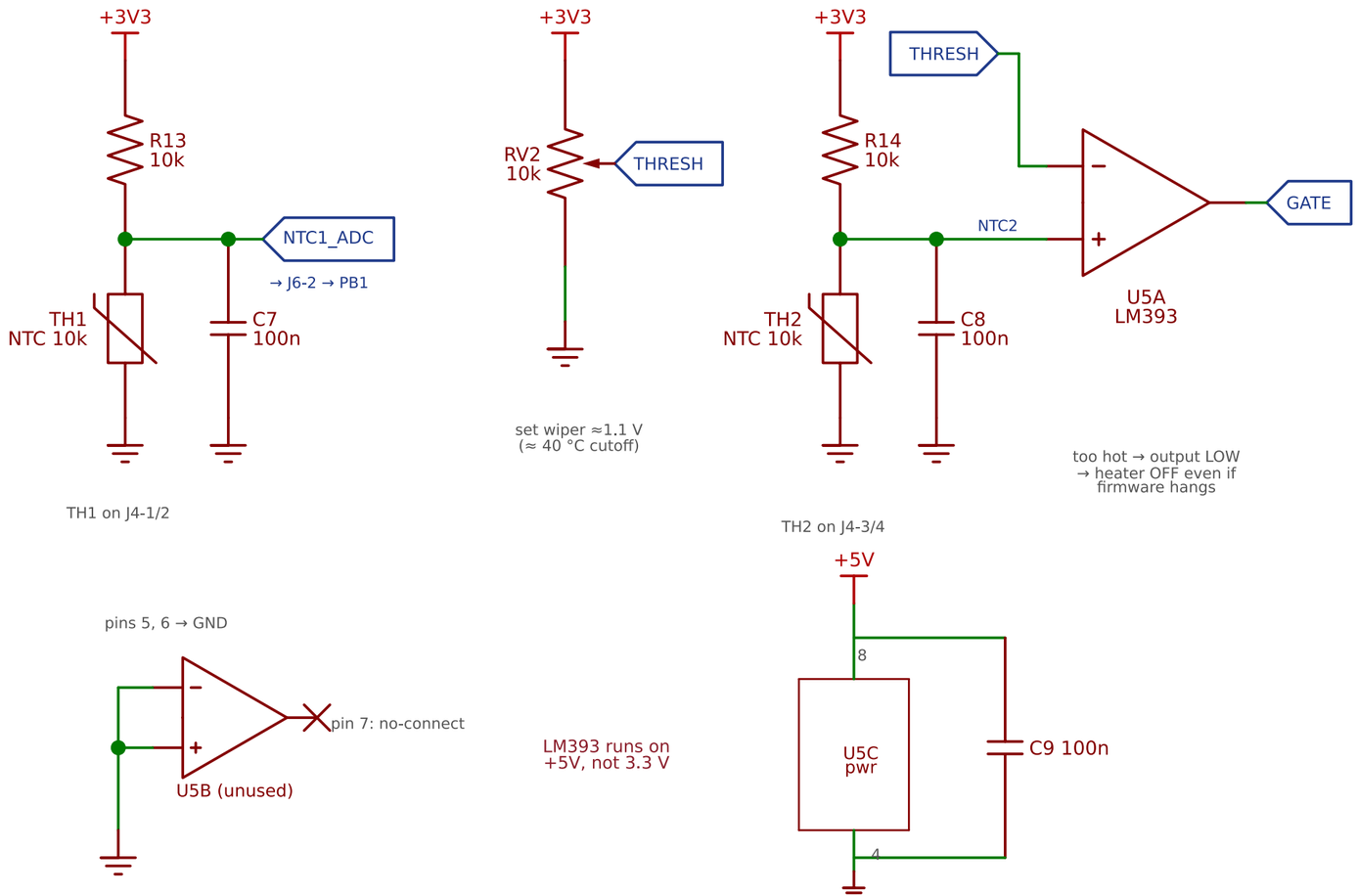

## B · 1.65 V reference (U2B) + U2 power




## C · Heater switch (14.8 V)




## D · Thermistors + hardware overheat cutoff (U5)




## E · Connectors — every pin and where it goes (same net name = connected)

### J5 power from FC

| | | |
|---|---|---|
| 1 | +5V | FC 5V |
| 2 | +3V3 | FC 3V3 |
| 3 | GND | FC GND |

### J2 XT30, 14.8 V from X8

| | | |
|---|---|---|
| 1 | +14V8 | X8 + |
| 2 | GNDPWR | X8 − |

### J6 to flight computer

| | | |
|---|---|---|
| 1 | HB_ADC | PB0 (ADC) |
| 2 | NTC1_ADC | PB1 (ADC) |
| 3 | PWM | PC6 (TIM PWM) |
| 4 | SCL | PD12 |
| 5 | SDA | PD13 |

### J4 thermistor cable

| | | |
|---|---|---|
| 1 | NTC1_ADC | TH1 |
| 2 | GND | TH1 |
| 3 | NTC2 | TH2 |
| 4 | GND | TH2 |

### J7 SHT21 module

| | | |
|---|---|---|
| 1 | +3V3 | VIN |
| 2 | GND | GND |
| 3 | SCL | SCL |
| 4 | SDA | SDA |

### J8 BMP280 module

| | | |
|---|---|---|
| 1 | +3V3 | VCC |
| 2 | GND | GND |
| 3 | SCL | SCL |
| 4 | SDA | SDA |
| 5 | GND | SDO (0x76) |

### J1 TCRT5000 cable

| | | |
|---|---|---|
| 1 | LED_A | LED anode |
| 2 | GND | LED cathode |
| 3 | +3V3 | collector |
| 4 | SENSE | emitter |

### J3 heater cable

| | | |
|---|---|---|
| 1 | +14V8 | R8–R10 top |
| 2 | HEATER_LO | R8–R10 bottom |

### J9 MAX30100 module

| | | |
|---|---|---|
| 1 | +3V3 | VIN |
| 2 | GND | GND |
| 3 | SCL | SCL |
| 4 | SDA | SDA |

+3V3 — C10 10µ (bulk on 3.3 V)